END TERM EXAMINATION

(CBCS)(SUBJECTIVE TYPE)

B. Tech CSE-AI , 2nd Semester

(AUGUST, 2022)

Subject Code:BAI 104	Subject: Introduction To Data Science

Time :3 Hours **Maximum Marks :60**

Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1	Write the output for the following program with explanation in two to three lines.	**(2.5*8=20)**

	(a) import pandas as pd
	def abc(n):
	  if n < 0:
	  return 0
	return n
	arr = pd.Series([-1, 0, 1])
	print(abc(arr))	

	(b) import pandas as pd s = pd.Series(['abc', 'def', 'ghi', 'jkl', 'mno']) print('abc' in s)	
	(c) import pandas as pd s = pd.Series(['Abc', 'Def', 'Ghi', 'Jkl', 'Mno']) print(s.lower())	
	(d)import pandas as pd grades = pd.Series([61, 82, 57]) bonuses = pd.Series([10, 5, 10, 10]) out = grades + bonuses print(out)	
	(e) import pandas as pd nums = pd.Series([1, 2, 3, 4, 5, 6]) print(nums[(nums> 2) and (nums< 5)])	
	(f) import numpy as np import pandas as pd s = pd.Series([1, 2, np.nan, 4, 5]) s.fillna(3) print(s.sum())	
	(g) import pandas as pd df1 = pd.DataFrame([[1, 2], [3, 4]], columns=['a', 'b']) df2 = pd.DataFrame([[5, 6], [7, 8]], columns=['b', 'c']) df = pd.concat([df1, df2]) print(df.dtypes)	

	(h)import pandas as pd
	df1 = pd.DataFrame({
	'id': [1, 2, 3],
	'name': ['Clark Kent', 'Diana Prince', 'Bruce Wayne'],
	})
	df2 = pd.DataFrame({
	'id': [2, 1, 4],
	'hero': ['Wonder Woman', 'Superman', 'Aquaman'],
	})
	df = pd.merge(df1, df2, on='id')
	print(df)	

	UNIT-I	
Q2	Find outliers usinginterquartile range IQR. Lower fence = Q1 – 1.5 (IQR) Upper fence = Q3 + 1.5 (IQR) Consider a sample dataset oftemperatures from a place at different times: 71, 70, 90, 70, 70, 60, 70, 72, 72, 320, 71, 69 Show the workingsteps for calculating outliers	**(10)**
Q3	(a)If a numeric variable is not normal, then by what value will it impute(replace missing value)?Explain with reason?. Ifvariable is categorical ,then by what value will it replace the missing value. (b) There are missing values and outlierin dataset. You have to impute missing values with mean of variable. State true or false for the following statement with reason.(First we will impute then will remove outlier)	**(10)**
	UNIT-II	
Q4	(a)Write an equivalent pandas program forthe following code without the pivot function pd.pivot_table(df, index='Post', columns='Project', values='Salary', aggfunc='sum') (b)Write data frame(df) to excel for each year data in separate sheet of excel file.	**(10)**
Q5	Write program using NumPy (a)Read dataset from csv file (b) Calculate the mean for the first column (c) Selecting every second element of the fifth row (d) Values that are between 90 and 95 (e) Sort values in descending order for each column	**(1+2+3+2+2)**
	UNIT-III	
Q6	IGDTUW is maintaining a database(df) of student with details like roll_no, name, height, weight,total_marks.....etc (a)Write a pandas or matplotlib code to plot a suitable chart to find if thereis any association between height and weight. (b)Write a pandas or matplotlib code to highlight with red color if marks is greater than 80 else green.	**(10)**
Q7	IGDTUW is maintaining a database (df) of student details like roll_no, name, height, weight, marks .....etc . (a)Write a pandas or matplotlib code to set background color with red color if marks is greater than 80 or else green. (b)Write a pandas or matplotlib code to plot a suitable chart to find the outlier in marks.	**(10)**
	UNIT-IV	
Q8	ABC online Superstore is maintaining a database named "retail" for daily transaction detail like order id,product _id ,segment,order_id, profit .....etc . In other database named "return" those products are returned(order_id ,product_id) .Write a code in Tableau for a calculated field percentage ofproduct returned.	**(10)**
Q9	ABC online Superstore is maintaining a database named retail for daily transaction details like product _id ,segment,order_id, profit, sales ..... .etc. Write a tableau code to create a calculated field for following condition. If sum of profit > 0 then create a label profit or else will create a label loss.	**(10)**